

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of Bachelor of Technology

University Examination May 2011

CE207 Java Programming

Date: 10.05.11, Tuesday

Time: 10.00 a.m to 1.00 p.m

Maximum Marks: 70

Instructions:

1. Attempt all questions.
2. Answers both sections in separate answer books.
3. Figure to the right indicates full marks.
4. Assume suitable data, if necessary and mention it.

Q-1 Do as Directed.

11

State True or False with justification.

1. A static method can't be overridden in Java.
2. A valid Java arithmetic expression with no parentheses is evaluated from left to right.
3. Primitive-type local variables are initialized by default.
4. Floating-point values that appear in source code are known as floating-point literals and are type float by default.
5. An array declaration reserves space for the array.

Fill in the Blanks.

6. All of the integer types except _____ are signed integers.
7. When each object of a class maintains its own copy of an attribute, the field that represents the attribute is also known as a(n) _____.
8. An object's _____ method is called implicitly when an object appears in code where a String is needed.
9. _____ keyword specifies that a variable is not modifiable.
10. _____ is designed to handle infrequently occurring situations that often result in program termination; the situations may not arise all the time.
11. The _____ statement selects among multiple actions based on the possible values of an integer variable or expression.

Q-2

- (a) Explain the different forms of Polymorphism by taking suitable example. 3
- (b) Which two conditions should be met so that an automatic type conversion will take place? Explain widening conversion and narrowing conversion. 6
- (c) How I/O is performed in java? Write a simple code which copies content from one binary file to another binary file. 3

OR

Q-2

- (a) Write down the differences between method overloading and method overriding. 3
- (b) List the logical operators are available in java. Explain how the short circuit logical operators (&&, ||) useful in programming? 6
- (c) How and why PATH and CLASSPATH variable have to be set to compile and run java program? 3

Q-3

- (a) What is an interface? Can we instantiate an interface? Which kind of member variables do interfaces have? 6
- (b) How are this() and super() used with constructors? Explain the multiple and multi level inheritance. Can super() be a last statement in constructor? Why? 6

OR

Q-3

- (a) What is an abstract class? Can we instantiate an abstract class? What are the differences between interface and abstract class? 6
- (b) What is Constructor? What are the differences between Constructor and Methods? If constructor is private in a class how that class can be instantiated? 6

Section II

Q-4 Do as Directed.

11

State True or False with justification.

1. Recursion can be efficient in computation because of reduced memory-space usage.
2. Applet can access local file system.
3. All methods in an abstract class must be declared as abstract methods.
4. When the mouse cursor is moved into the bounds of a GUI component, method mouseOver is called.
5. The first two arguments of Graphics method drawOval specify the center coordinate of the oval.

Fill in the Blanks.

6. The method _____ is invoked for an applet each time the user of a browser leaves an HTML page on which the applet resides.
7. The class _____ is used to create a menu object.
8. A Runnable thread can enter the _____ state for a specified interval of time.
9. The statement _____ indicates that only one thread at a time should execute on an object.
10. The file produced by the Java compiler contains _____ that are executed by the Java Virtual Machine.
11. The add method for attaching GUI components is a method of class _____.

Q-5

- (a) Which types of access specifiers available in Java? Explain how final modifier can be affect to declaration of class, method and data? 6
- (b) What is static block? What are static variables? Mention the difference between static and non-static variables? 6

OR

Q-5

- (a) Explain exception handling. Can we have the try block without catch block? What is the difference between throw and throws? 6
- (b) How thread is useful in programming? List the possible ways to create threads. Differentiate setDeamon(boolean) and join(int) methods. 6

Q-6 Write a short note on any three:

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- (a) Life cycle of an applet
- (b) Layout managers
- (c) Event handling using AWT
- (d) Types of classes in java